

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2011

EC 301 Electromagnetic Theory

Date: 12.12.2011, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Illustrating neat diagrams perform conversion of cylindrical coordinate system to Cartesian coordinate system and vice versa. [04]
- (b) Given the points $M(0.1, -0.2, -0.1)$, $N(-0.2, 0.1, 0.3)$, and $P(0.4, 0, 0.1)$, calculate [03]
(1) the vector R_{MN} , (2) the dot product of two vectors $R_{MN} \cdot R_{MP}$ (3) the scalar projection of vector R_{MN} on vector R_{MP}
- Q - 2 (a) Derive Boundary conditions for perfect dielectric materials [07]
- (b) By establishing dipole moment, explain magnetization and permeability using [07]
equations. In addition, derive relationship between $\vec{B}$ and $\vec{H}$.

OR

- Q - 2 (a) What is the condition of existence of Laplace equation? State and prove uniqueness theorem. [07]
- (b) Let $\vec{J} = 400 \sin \theta / (r^2 + 4) \vec{a}_r \text{ A/m}^2$. (1) Find the total current flowing through that portion of the spherical surface $r = 0.8$, bounded by $0.1\pi < \theta < 0.3\pi$, $0 < \phi < 2\pi$. (2) Find the average value of $\vec{J}$ over the defined area. [07]
- Q - 3 (a) Mention significance of cross product. Give example illustrating the application of Faraday's law in the case of moving shorting bar of a constant magnetic flux density $\vec{B}$. [07]
- (b) What is meaning of retarded potentials. Derive equation of retarded vector magnetic potential. [07]

OR

- Q - 3 (a) Derive any two Maxwell's equations in integral form. [07]
- (b) Using neat diagrams explain concept of current density and relation of current and current density. [07]

SECTION – II

Q - 4 (a) Find $\nabla \times \vec{H}$ if $\vec{H}$ equals [04]

(1) $y^2 z \vec{a}_x + 2(x+1)yz \vec{a}_y - (x+1)z^2 \vec{a}_z$

(2) $2\rho \cos \phi \vec{a}_\rho - 4\rho \sin \phi \vec{a}_\phi + 3\vec{a}_z$

(b) A dipole in free space is formed by a charge of 1 nc at (0,0,0.1) and -1nc at (0,0,-0.1). [03]

At point p($r = 0.2$, $\theta = \pi/4$, $\phi = 0$). Calculate $|\vec{E}|$

Q - 5 (a) State the divergence theorem. Derive the point form of ampere circuital law. [07]

(b) Briefly differentiate dot product and cross product. Prove that [07]

$$\nabla \cdot (VD) = D \cdot \nabla V + V(\nabla \cdot D)$$

OR

Q - 5 (a) Define potential difference. Derive electric flux density due to co-axial cable. [07]

(b) Define Curl. Prove: $\nabla \times \nabla \times \vec{A} = \nabla(\nabla \cdot \vec{A}) - (\nabla^2 \vec{A})$ [07]

Q - 6 (a) Define TEM Wave and Explain Wave Propagation in Free Space. In addition, derive intrinsic impedance of free space. [07]

(b) With help of necessary equations and diagrams, explain Standing Wave Ratio and its significance. [07]

OR

Q - 6 (a) Explain Poynting's Theorem and Wave Power with use of necessary equations and give its significance. [07]

(b) Find the incremental contribution $\Delta \vec{H}$ to the magnetic field intensity caused by a current element in free space $I \Delta L$ equal to (1) $3\pi \vec{a}_z \mu A \cdot m$ located at (3,-4,0) (2) $3\pi \vec{a}_z \mu A \cdot m$ located at (3,2,-4) at origin for both cases. [07]
